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CDN Cannon: Exploiting CDN Back-to-Origin Strategies for Amplification Attacks

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Content Delivery Networks (CDNs) provide high availability, speed up content delivery, and safeguard against DDoS attacks for their hosting websites. To achieve the aforementioned objectives, CDN designs several 'back-to-origin' strategies that proactively pre-pull resources and modify HTTP requests and responses. However, our research reveals that these 'back-to-origin' strategies prioritize performance over security, which can lead to excessive consumption of the website's bandwidth.

We have proposed a new class of amplification attacks called **Back-to-Origin Amplification (BtOAmp)** Attacks. These attacks allow malicious attackers to exploit the 'back-to-origin' strategies, triggering the CDN to greedily demand more-than-necessary resources from websites, which finally blows the websites. We evaluated the feasibility and real-world impacts of 'BtOAmp' attacks on fourteen popular CDNs. With real-world threat evaluation, our attack threatens all mainstream websites hosted on CDNs. We responsibly disclosed the details of our attack to the affected CDN vendors and proposed possible mitigation solutions.

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